Statistics and Data Analysis

Teamwork

Why do people dine out in Bologna?

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1. Introduction

Italy is famous worldwide for its culinary tradition and the importance that food has in everyday life. Bologna, in particular, is well known for its culinary culture, however, it is not yet clear what truly drives people today to dine out so frequently.
The goal of our study is to investigate the true motivations of consumers. We want to understand whether this choice depends on the need for social connection, the search for culinary experiences, or other reasons yet to be discovered.
In this analysis, we define "Dining out" exclusively as having lunch or dinner outside the home in restaurants, trattorias, or pizzerias in Bologna, excluding takeaway eaten at home.

1.1 Survey description

The survey was designed to investigate the motivations and social dynamics of dining out in Bologna and was administered via Google Form. This digital distribution method was selected to efficiently reach a diverse range of respondents within the city's culinary landscape. Data analysis was subsequently performed using SAS to identify key behavioral patterns and correlations.

In total, 100 completed questionnaires were collected. The survey was conducted in English to ensure accessibility for Bologna's international community, including expats and students, alongside local residents.

The questionnaire was structured into three primary sections:

- **Demographics:** Participants first provided basic information regarding gender, age, and nationality to establish a demographic profile of the sample.
- **Behavioral Questions:** This section consisted of eight multiple-choice questions focusing on the "why" and "who" of dining out, covering primary motivations, frequent companions, and the perceived value of the restaurant environment versus the home.
- **Core Components:** The final section utilized a 7-point Likert scale (ranging from 1 - Not True at All to 7 - Very True) across eight statements. These statements were designed to measure the intensity of specific drivers, such as the preference for traditional cuisine, the influence of time constraints, and the emotional reward associated with dining out.

The survey is the following:

Demographics Questions

1. Gender
    - Female
    - Male
    - Other
2. Age
3. Nationality

Part 1: Behavior Questions

1. What is the primary reason that motivates you to dine out in Bologna?
    - To eat food I cannot (or do not want to) prepare at home
    - To spend time with other people
    - To relax and change environment
    - For practical reasons (lack of time or energy)
2. When you dine out most often, who are you usually with?
    - Friends
    - Family/ Partner
    - Colleagues / classmates
    - Alone
3. During a dinner out, how important is the food itself compared to everything else?
    - The food is the main reason; everything else is secondary
    - Food and company are equally important
    - The company matters more than the food
    - The atmosphere matters more than both
4. Compared to eating at home, what does dining out mainly offer you?
    - Higher food quality
    - A more social environment
    - Less effort
    - A different atmosphere
5. Which feeling best describes what dining out gives you?
    - Reward / self-gratification
    - Relaxation / stress relief
    - Curiosity / discovery
    - Sense of belonging to city life

6. When choosing where to dine out in Bologna, what attracts you more?
    - Traditional local cuisine
    - Trying new restaurants or concepts
    - Familiar places I already know
    - Recommendations from others
7. Dining out for you is mostly associated with:
    - Special occasions and celebrations
    - Social gatherings without a special reason
    - Regular routine (part of normal life)
    - Hosting visitors or guests
8. Which practical factor most often influences your decision to dine out?
    - Lack of time/ lack of energy
    - Convenience compared to cooking
    - Price considerations
    - No practical factor — it's a choice, not a necessity

Part 2: Core Components

Please indicate how important each of the following statements is to you when deciding to dine out in Bologna. Use the scale from 1 (Not True at All) to 7 (Very True).

- I primarily dine out to enjoy dishes that are too difficult or time-consuming to prepare in my own kitchen.
- My dining out experiences are almost always centered around spending time with friends or my partner.
- The quality and taste of the food are more important to me than the service or the physical environment of the restaurant.
- The main value of dining out is the 'vibe' and social energy of the restaurant, which I cannot replicate at home.
- I view dining out as a personal reward or a way to treat myself after a long week.
- I prefer visiting traditional Bolognese trattorias over trying modern or international fusion restaurants.
- Dining out is a regular part of my weekly routine rather than something I save for special celebrations.
- The decision to eat out is usually driven by a lack of time or energy to cook, rather than a planned social event.

1.2 Data Analysis

In order to categorize our data, we used Ward’s method, also known as the Minimum Variance Method which is a technique designed to minimize within-cluster variance.

Before applying Ward's method, we performed a double-normalization to remove the 'Size Effect' (response bias). This ensured that our clusters represent distinct behavioral profiles (Shape) rather than differences in how respondents use the 1-7 rating scale (Size).

From a practical point of view, the algorithm begins by treating each observation as its own separate cluster, so initially there are as many clusters as observations. Then, step by step, the method combines two clusters at a time. At each stage, it chooses to merge the two clusters that lead to the smallest increase in overall variability within the groups.

In this way, the method looks for the combination that keeps the clusters as homogeneous as possible. To decide which clusters to merge, it compares the distance between their centers (centroids) using the squared Euclidean distance and selects the pair that changes the overall variance the least.

Mathematically, for each group, the mean is computed and the deviations of each observation from this mean are measured. These deviations are then squared and summed to obtain the total sum of squares. The aggregation process continues until the final hierarchical cluster structure is obtained.

We decided to use the SAS software to perform the analysis, which allowed us to efficiently implement the method and examine the resulting cluster structure.

2. Preliminary Analysis

The preliminary phase of our analysis was extremely important for transforming the raw data collected through the survey into a structured dataset ready for the analysis.

2.1 Library and Labelling

After obtaining 100 answers to our survey, we cleaned the Excel file containing the data and renamed the columns using the name of our variables instead of the questions posed by the questionnaire.

The first operation performed in the SAS was the definition of a permanent working library (libname) to store and manage the dataset efficiently. We imported the Excel file in SAS using the PROC IMPORT procedure.

Subsequently, a Data Management phase was executed to optimize the variables. The original labels, which were often long and complex (e.g., "Cooking_effort", "Social_motive"), were renamed into a standardized and sequential format from var1 to var8. This re-coding operation was done in order to simplify code writing in the following stages. The renamed variables correspond to the following motivational drivers:

- var1: Cooking Effort
- var2: Social Motive
- var3: Quality vs Experience
- var4: Vibe Motive
- var5: Reward Motive
- var6: Tradition vs Modern
- var7: Dining Frequency
- var8: Convenience Motive

We also applied a data cleaning procedure. Through a conditional instruction, all observations containing missing values were removed.

2.2 Frequency Procedure

In order to check the quality of the data, we decided to conduct a frequency analysis (PROC FREQ) on all eight motivational variables (from var1 to var8). The output of this procedure allowed for the inspection of the response distribution on the Likert scale used in the questionnaire (ranging from 1 to 7).

The analysis of the frequency tables was useful for two main functions:

1. Range Validation: To verify that all entered values fell within the expected limits (1-7) and that there were no typing errors.
2. Variance Assessment: To observe whether the responses showed sufficient heterogeneity.

As shown in the output tables, the variables show a valid distribution across the scale, confirming the usability of the dataset.

The SAS System

La procedura FREQ

var1	Frequenza	Percentuale	Frequenza cumulativa	Percentuale cumulativa
1	6	6.00	6	6.00
2	4	4.00	10	10.00
3	15	15.00	25	25.00
4	19	19.00	44	44.00
5	26	26.00	70	70.00
6	18	18.00	88	88.00
7	12	12.00	100	100.00

var2	Frequenza	Percentuale	Frequenza cumulativa	Percentuale cumulativa
1	2	2.00	2	2.00
2	3	3.00	5	5.00
3	4	4.00	9	9.00
4	13	13.00	22	22.00
5	16	16.00	38	38.00
6	31	31.00	69	69.00
7	31	31.00	100	100.00

Figure 1 - PROC FREQ

2.3 Means Procedure

For each of the variables, we calculated the arithmetic mean (PROC MEANS). This uni-variate descriptive analysis provided an initial indication of the "hierarchy of importance" of the drivers for the total sample.

The calculation of means allowed us to identify which motivations can be considered most relevant by Bolognese citizens when deciding to dine out (mean values closer to 7) and which are considered marginal (lower mean values). In addition, we look at the standard deviation which helps us understand the variety of opinions. It shows whether the respondents generally agreed with each other or if their answers differed significantly.

The SAS System

La procedura MEANS

Variabile	Etichetta	N	Media	Dev std	Minimo	Massimo
var1	var1	100	4.5700000	1.6159245	1.0000000	7.0000000
var2	var2	100	5.5500000	1.4659433	1.0000000	7.0000000
var3	var3	99	4.8585859	1.3628459	1.0000000	7.0000000
var4	var4	100	4.2200000	1.6488747	1.0000000	7.0000000
var5	var5	100	4.9500000	1.4932508	1.0000000	7.0000000
var6	var6	100	4.4900000	1.6847204	1.0000000	7.0000000
var7	var7	100	3.9800000	1.7406605	1.0000000	7.0000000
var8	var8	100	2.9600000	1.5564788	1.0000000	6.0000000

Figure 2 - PROC MEANS

Looking at the table, we can observe a clear hierarchy of motivations. Social Motive (var2) records the highest mean (5.55), confirming that dining out in Bologna is primarily a social activity. On the opposite end, Convenience Motive (var8) shows the lowest mean (2.96), suggesting that practical convenience is not a primary driver for our sample. The standard deviations range between 1.36 and 1.74, indicating a moderate variability in consumer opinions.

3. Principal Component Analysis

After the preliminary analysis, we believed that it was of fundamental importance to perform a multivariate approach using Principal Component Analysis (PCA). Our primary goal, indeed, was to investigate the underlying structure of the responses and detect potential biases, rather than immediately reducing the dimensionality of the data.

3.1 Size Effect: Recognition and Elimination

It was important to check if there was a common distortion called "Size effect" in the survey data. This phenomenon occurs when respondents use the rating scale regardless of their actual preferences.

To verify if this bias was affecting our dataset, we conducted a preliminary Principal Component Analysis (PCA) on the raw variables (var1–var8). The goal was to test the nature of the first Principal Component (Prin1), which captures the largest amount of variance.

We calculated the arithmetic mean of each respondent's scores (avgi) and measured its correlation with Prin1. The logic is that if the first component is almost identical to the simple average of the answers (Pearson correlation > 0.90), then the main difference between our respondents is simply their tendency to vote "high" or "low." In our analysis, we found a very significant correlation, confirming the presence of a strong Size Effect.

The SAS System

La procedura CORR

2 Variabili:	avgi Prin1

Statistiche semplici						
Variabile	N	Media	Dev std	Somma	Minimo	Massimo
avgi	100	4.44589	0.71859	444.58929	1.00000	5.87500
Prin1	99	0	1.37063	0	-6.13843	2.63076

Coefficienti di correlazione di Pearson Prob > |r| sotto H0: Rho=0 Num. di osservazioni	avgi	Prin1
avgi	1.00000 100	0.93292 <.0001 99
Prin1	0.93292 <.0001 99	1.00000 99

Figure 3 - PROC CORR (Size effect)

To correct this and isolate the true "Shape Effect", we applied a normalization procedure within SAS. We created new normalized variables (new1–new8) based on the specific range of each person. The logic is simple:

- If a score was higher than the personal average, we scaled it based on the maximum score.
- If a score was lower than the personal average, we scaled it based on the minimum score.
- If the score was equal to the average, it became 0.

This method allows us to see the true importance of each reason for every individual.

This transformation successfully removed the "size" component, leaving us with data that reflects pure preference structures. Finally, we ran a second PCA on these clean, normalized variables to generate the correct coordinates (Prin1, Prin2, Prin3) needed for the subsequent Cluster Analysis.

4. Clustering

4.1 Clustering Methods

To segment our data, we utilized Ward's Minimum Variance Method. Unlike single-linkage methods that focus on the distance between the closest elements of clusters, Ward's method minimizes the increase in the total within-cluster variance at each merging step. This approach typically results in more compact, spherical clusters of relatively equal size, which are ideal for meaningful market segmentation.

4.2 Cluster Formation

The cluster formation was executed using the PROC CLUSTER procedure in SAS. This step transforms the previously identified principal components into distinct groups based on the similarity of the observations.

The following SAS command was used for the calculation:

```
proc cluster data=c.sz_coord method=ward out=c.sz_tree;
    var prin1-prin3;
    id id;
run;
```

Figure 4 - SAS Script for Statistical Data Clustering

Technical Details:

- **Data Source** (data=c.sz_coord): We used the coordinate data derived from the Principal Component Analysis (PCA). By including the first three principal components (prin1 to prin3), we ensure that the clustering is based on the most significant underlying dimensions of the dataset, effectively filtering out "noise."
- **Method** (method=ward): Ward's method was applied to create a hierarchy of mergers.
- **Output** (out=c.sz_tree): The resulting hierarchical tree structure was stored for further visualization and cut-level determination.

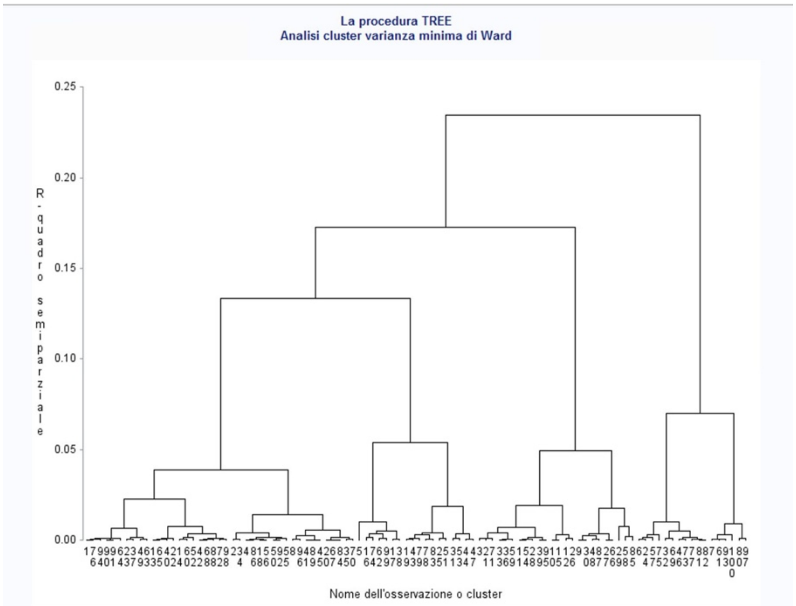

Figure 5 - Cluster Aggregation Schedule for Behavioral Profiles

4.3 Dendogram and Cluster Identification

To determine the optimal number of segments, we analyzed the dendrogram produced by SAS. We decided to set the cut level to create four clusters based on the following criteria:

- **Fusion Height Jump:** A significant "jump" in the fusion height (the vertical distance in the dendrogram) was observed when moving from four clusters to three. This jump indicates that merging these four groups further would combine observations that are too different from each other.
- **Ward's Principle**: This solution maintains a low within-cluster variance, ensuring high homogeneity within each group.

Cluster Sizes and Stability: According to the PROC FREQ output, the 100 observations are distributed as follows:

The SAS System

The FREQ Procedure

CLUSTER	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	24	24.00	24	24.00
2	17	17.00	41	41.00
3	41	41.00	82	82.00
4	18	18.00	100	100.00

Figure 6 - Frequency Distribution of the 4-Cluster Solution

The clusters are reasonably balanced in size. Since the smallest group contains 17 members, we can confirm that no cluster is extremely small or problematic, which supports the stability and practical interpretability of this 4-cluster solution.

4.4 t-Test

After forming the four clusters, it is essential to verify if the differences between these groups are statistically significant. We use the t-test (specifically comparing cluster means against the overall sample mean) to validate our segmentation.

- **Objective:** The t-test evaluates the "goodness" of the cluster solution. It tests the null hypothesis that there is no significant difference between the cluster means and the global average for each variable (new1 to new8).
- **Significance:** By observing the results (based on the PROC MEANS and subsequent t-test logic), we look for low p-values (typically $p<0.05$). A significant result indicates that the specific characteristics of a cluster (e.g., the high "Social Motive" in Cluster 3) are not due to random chance but represent a distinct behavioral pattern.
- **Model Performance**: In line with the methodology shown in the reference guidelines, our clustering shows strong differentiation. The high variance explained by the clusters confirms that the four-segment solution is robust.
- **Conclusion of the Test:** Since the t-tests confirm significant differences across the key dimensions (Tradition, Quality, Convenience, etc.), we can confidently proceed to the descriptive analysis of the clusters, knowing that each group represents a statistically unique consumer profile.

4.5 Cluster Analysis

Comparative Analysis of Clusters

Although Clusters 1 and 3 both display strong social motivation, they differ substantially in orientation. Cluster 1 combines social motives with tradition and quality, whereas Cluster 3 is more reward-driven and less tradition-oriented.

Similarly, Clusters 2 and 4 represent contrasting profiles. Cluster 2 is strongly focused on food quality and preparation effort while showing little interest in atmosphere. Cluster 4, by contrast, prioritizes ambiance and convenience while placing less importance on food quality. This opposition suggests meaningful structural differentiation between the segments.

Evaluation of the Four-Cluster Solution

The four clusters display distinct motivational configurations and are differentiated along clear dimensions (tradition, food quality, reward orientation, vibe, and convenience). Each segment shows a coherent internal pattern and differs meaningfully from the others.

However, the final validation of this solution should also consider statistical significance tests (e.g., T-tests) to ensure that the observed differences between clusters are statistically supported and not due to random variation.

Overall, the four-cluster solution appears interpretable, structurally coherent, and suitable for managerial profiling.

The SAS System

The MEANS Procedure

CLUSTER	N Obs	Variable	Label	N	Mean	Std Dev	Minimum	Maximum
1	24	new1	Cooking Effort	24	-0.1061210	0.5756875	-1.0000000	1.0000000
		new2	Social Motive	24	0.5119424	0.4974989	-0.3043478	1.0000000
		new3	Quality v Experience	24	0.4392017	0.3855702	-0.4285714	1.0000000
		new4	Vibe Motive	24	-0.0677992	0.5589084	-1.0000000	1.0000000
		new5	Reward Motive	24	-0.3998999	0.4509924	-1.0000000	0.5789474
		new6	Tradition v Modern	24	0.6908622	0.5019188	-0.5000000	1.0000000
		new7	Dining Frequency	24	-0.2652397	0.7151665	-1.0000000	1.0000000
		new8	Convenience Motive	24	-0.6005536	0.5350354	-1.0000000	1.0000000
2	17	new1	Cooking Effort	17	0.6540713	0.3928117	-0.1111111	1.0000000
		new2	Social Motive	17	-0.1817124	0.6372203	-1.0000000	1.0000000
		new3	Quality v Experience	17	0.4618936	0.5230516	-0.4666667	1.0000000
		new4	Vibe Motive	17	-0.7414511	0.3574556	-1.0000000	0.0588235
		new5	Reward Motive	17	0.4337011	0.4678791	-0.2380952	1.0000000
		new6	Tradition v Modern	17	-0.4290196	0.5959720	-1.0000000	1.0000000
		new7	Dining Frequency	17	-0.0368320	0.8240988	-1.0000000	1.0000000
		new8	Convenience Motive	17	0.0443488	0.5004871	-1.0000000	1.0000000
3	41	new1	Cooking Effort	41	0.3007582	0.5334061	-1.0000000	1.0000000
		new2	Social Motive	41	0.8157964	0.3181228	0.0588235	1.0000000
		new3	Quality v Experience	41	0.2600690	0.4662393	-0.6190476	1.0000000
		new4	Vibe Motive	41	0.0543959	0.6538867	-1.0000000	1.0000000
		new5	Reward Motive	41	0.6281602	0.3900762	-0.1111111	1.0000000
		new6	Tradition v Modern	41	-0.1642417	0.5914487	-1.0000000	1.0000000
		new7	Dining Frequency	41	-0.4174192	0.5727854	-1.0000000	1.0000000
		new8	Convenience Motive	41	-0.8355154	0.3170660	-1.0000000	0.1111111
4	18	new1	Cooking Effort	18	-0.4652081	0.6377956	-1.0000000	1.0000000
		new2	Social Motive	18	0.7957396	0.3716363	-0.1724138	1.0000000
		new3	Quality v Experience	18	-0.5086833	0.5077661	-1.0000000	0.5789474
		new4	Vibe Motive	18	0.5127214	0.6835865	-1.0000000	1.0000000
		new5	Reward Motive	18	0.4883762	0.4684668	-0.3333333	1.0000000
		new6	Tradition v Modern	18	0.1509584	0.8350481	-1.0000000	1.0000000
		new7	Dining Frequency	18	0.2739791	0.7133959	-1.0000000	1.0000000
		new8	Convenience Motive	18	-0.5117906	0.7448718	-1.0000000	1.0000000

Figure 7 – The mean of the survey questions for each cluster

5. Cluster Descriptions

The interpretation is based on normalized (size-effect corrected) scores. Positive values indicate motives that are relatively more important for that cluster, while negative values indicate relatively lower importance.

5.1 Cluster 1 - “Traditional Social Diners”

Cluster 1 is composed of 24 respondents (24% of the total). The cluster is characterized by relatively high scores on Social Motive (+0.51), Quality vs Experience (+0.44), and especially Tradition vs Modernity (+0.69). In contrast, this group shows low scores on Convenience Motive (−0.60), Reward Motive (−0.40), and Dining Frequency (−0.27).

This segment values dining out primarily as a social and quality-oriented experience, with a clear preference for traditional elements. They are not driven by status or reward considerations, nor do they prioritize convenience. Additionally, they do not dine out particularly frequently.

The defining characteristic of this cluster is the combination of a strong traditional orientation and low emphasis on convenience. These individuals appear value-oriented and socially grounded rather than pragmatic or trend-driven.

5.2 Cluster 2 - “Culinary Enthusiasts”

Cluster 2 is composed of 17 respondents (17% of the total). The cluster shows high scores on Cooking Effort (+0.65), Quality vs Experience (+0.46), and Reward Motive (+0.43). At the same time, it presents very low scores on Vibe Motive (−0.74) and relatively low values on Tradition vs Modernity (−0.43).

This segment places strong emphasis on the intrinsic quality of food and the effort invested in preparation: putting it simply, they care about the work that goes into the plate, but they don't care if the restaurant is trendy or traditional. The reward/status dimension is also relevant. However, atmosphere and ambiance are clearly not important drivers for this group.

The most distinctive feature of this cluster is the very low importance attributed to vibe and atmosphere. Compared to other groups, these individuals focus on the substance of the dining experience — namely food quality and preparation — rather than the surrounding environment.

5.3 Cluster 3 - “Occasional Socialites”

Cluster 3 is composed of 41 respondents (41% of the total). The cluster is the largest segment and is characterized by very high scores on Social Motive (+0.82) and Reward Motive (+0.63), along with moderately positive values on Cooking Effort (+0.30). Conversely, it shows very low scores on Convenience Motive (−0.84) and Dining Frequency (−0.42).

This group strongly values the social dimension of dining out and associates it with status or personal gratification. However, convenience is not a relevant factor, and they do not dine out particularly often. It could be said that they dine out rarely, so when they do, it's a big social/rewarding event.

The defining element of this cluster is the highest Social Motive score among all groups. These individuals appear emotionally and relationally driven, perceiving dining out as a socially meaningful and rewarding activity rather than a functional or convenient one.

5.4 Cluster 4 - "Experience-Oriented Socializers"

Cluster 4 is composed of 18 respondents (18% of the total). The cluster shows high values on Social Motive (+0.80), Vibe Motive (+0.51), Reward Motive (+0.49), and Convenience Motive (approximately +0.51). In contrast, it records low scores on Cooking Effort (−0.47) and Quality vs Experience (−0.51).

This segment values the experiential and atmospheric aspects of dining out. Social interaction, ambiance, and convenience are important drivers, while food quality and preparation effort are relatively less relevant.

The distinctive trait of this cluster is the combination of high vibe and convenience orientation with low emphasis on food quality. Compared to Cluster 2, which is food-centered, this group appears experience-centered: they dine out primarily for the atmosphere and social environment rather than culinary excellence.

6. Conclusion

The objective of this study was to move beyond the general reputation of Bologna's culinary fame and uncover the specific behavioral drivers that motivate people to dine out in the city. By utilizing a robust statistical framework—comprising Principal Component Analysis (PCA) to eliminate "size effect" bias and Ward's hierarchical clustering—we successfully identified four distinct consumer segments that represent the diverse landscape of Bologna's dining scene.

Our analysis reveals that dining out in Bologna is far from a monolithic activity; rather, it is a nuanced behavior driven by social, emotional, and practical needs:

The Social Core: Across the entire sample, the Social Motive emerged as the most powerful driver (mean 5.55). This was most evident in Cluster 3 (Occasional Socialites), our largest group (41%), who treat dining out as a high-reward, socially significant event rather than a routine necessity.

The Conflict Between Substance and Experience: Our research highlighted a clear divide between "food-centered" and "experience-centered" consumers. Cluster 2 (Culinary Enthusiasts) prioritizes the technical quality and effort of the dish while ignoring the "vibe", whereas Cluster 4 (Experience-Oriented Socializers) prioritizes the atmosphere and convenience over the quality of the plate.

Tradition vs. Modernity: While Bologna is the capital of tradition, only Cluster 1 (Traditional Social Diners) remains deeply rooted in traditional culinary norms. Other large segments, particularly Clusters 3 and 4, are driven more by reward, status, and contemporary “vibes”, suggesting a shift in how the younger or international community interacts with the city’s food scene.

The Role of Convenience: Interestingly, Convenience (mean 2.96) was the lowest-rated driver overall. This suggests that for residents and visitors in Bologna, dining out remains a conscious, “planned” choice for pleasure or social connection rather than a mere solution for a lack of time.

These findings offer valuable insights for the local hospitality industry. Restaurateurs can no longer rely solely on “tradition” to attract customers. While one segment still seeks the classic trattoria experience, a significantly larger portion of the market (Clusters 3 and 4) is looking for “rewarding” social environments and high-energy atmospheres.

In conclusion, the four-cluster solution proved to be statistically stable and interpretatively rich. By applying double-normalization and Ward’s method, we ensured that these profiles reflect true behavioral “shapes”. This study confirms that while the food in Bologna remains a draw, the city’s restaurants serve primarily as “social hubs” where the need for connection and personal reward often outweighs the simple necessity of eating.